

Final Consulting Report

Hospital Readmission Reduction Analytics Case Study:

Identifying Factors Associated with 30-Day Readmission and Opportunities for Intervention

Client: Regional Hospital System

Sponsor: Chief Quality Officer

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Analysis Period: January 2015 – December 2024

1. Business Objective

The project was designed to help the hospital use available patient and clinical data to:

- establish the current 30-day readmission baseline;
- identify populations and characteristics associated with readmission;
- distinguish high-rate populations from high-volume readmission populations;
- develop an exploratory patient-level risk-stratification approach; and
- translate the findings into potential intervention and monitoring priorities.

The detailed business discovery, questions, and data requirements are documented separately in the project repository:

[Consulting_Healthcare_Readmission_Reduction_Case_Study >> 02_Business Discovery](#)

2. Data & Analytical Approach

2.1 Data: The analysis period covers January 2015 through December 2024. A total of five datasets were assessed: patients.csv, admissions.csv, diagnoses.csv, billing.csv, and hospitals.csv.

2.2 Data Quality: Overall, the datasets demonstrate good data quality and are suitable for the proposed readmission analytics engagement. No duplicate records, primary key uniqueness issues, or referential integrity violations were identified. The primary data

quality issue was 24,715 missing values in patients.insurance_type, representing 3.58% of patient records. These records will be retained, with missing insurance values excluded from insurance-specific analyses where appropriate.

An additional hospital identification issue was identified: the hospitals dataset contains 33 unique hospital_id values but only 26 unique hospital names. This indicates that hospital names are not sufficient to uniquely distinguish all hospital entities. Some hospitals share the same name but have different bed counts. To ensure accurate hospital-level analysis and reporting, a derived hospital_name field combining the hospital name and bed count was created during the staging transformation.

It is worth noting that the 30-day readmission rate calculated among admissions discharged alive; 7,413 admissions with expired discharge were excluded because they were not eligible for subsequent readmission.

The admit_date and discharge_date fields in the admissions dataset require conversion from string to date/datetime format during SQL ETL. No other material data quality issues were identified

A detailed Data Inventory and Data Quality Assessment is provided separately in the project repository:

[Consulting_Healthcare_Readmission_Reduction_Case_Study >> 04_Data](#)

Technical Python work is documented in Github repository:

[Consulting_Healthcare_Readmission_Reduction_Case_Study >> 06_Python >> 06_01_Data_Profiling_and_Data_Quality.ipynb](#).

2.3 Analytical Approach: The analysis followed three complementary levels.

2.3.1 Descriptive and Comparative Analysis: SQL/BigQuery was used to establish the readmission baseline and examine differences across demographic, clinical, utilization, admission, discharge, and hospital characteristics.

This analysis addressed: “Where is the readmission burden concentrated, and which characteristics are associated with higher observed readmission rates?”

Technical SQL work is documented in Github Project Repository:

[Consulting_Healthcare_Readmission_Reduction_Case_Study >> 05_SQL_BigQuery](#)

2.3.2 Statistical Analysis: ANOVA was used for continuous variables and chi-square testing for categorical variables in the focused secondary-diagnosis analysis. Effect sizes were included to provide context on the magnitude of observed differences.

Technical Python work is documented in Github repository:

[Consulting_Healthcare_Readmission_Reduction_Case_Study >> 06_Python >> 06_02_Statistical_Analysis_Readmission_Burden.ipynb](#)

2.3.3 Predictive Modeling: Predictive models were developed to estimate 30-day readmission risk at the point of discharge, using information available by that point in the patient encounter. Logistic Regression, Random Forest, and XGBoost were evaluated using a held-out test set. Several Logistic Regression specifications were also examined to assess whether more detailed diagnosis representations provided meaningful improvement over a simpler feature set.

Model performance was evaluated primarily using test-set AUC. The final model was selected based on the combination of discrimination, feature complexity, and interpretability, consistent with the exploratory operational purpose of the analysis.

Detailed feature engineering, model development, and evaluation are documented in the project repository: [Consulting_Healthcare_Readmission_Reduction_Case_Study >> 06_Python >> 06_03_Readmission Predictive_Modeling.ipynb](#)

3. Findings & Insights

3.1 The Readmission Problem Is Persistent

The hospital recorded 120,000 admissions, of which 112,587 were discharged alive and therefore eligible for 30-day readmission. The overall 30-day readmission rate was 12.62%, with 14,210 30-day readmissions. In comparison, only 1,006 readmissions occurred within 7 days. This indicates that the majority of readmissions occurred after the first week of discharge, during days 8–30, highlighting the importance of extending transitional-care and follow-up efforts beyond the immediate post-discharge period.

At the population level, the 30-day readmission rate was relatively stable across the historical years and showed no substantial month-to-month variation in the available data. This suggests that the hospital's readmission challenge is persistent rather than driven by a particular month or year, supporting the need for sustained intervention rather than a short-term or seasonal response.

3.2 Where Is the Readmission Problem?

Dashboard 1: Link to Executive Dashboard at Tableau Public

https://public.tableau.com/views/HospitalReadmissionConsultationProject/Dashboard4?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

The analysis shows that readmission risk and readmission burden are not the same thing. Some groups have substantially higher readmission rates but represent relatively smaller populations, while other groups have more moderate rates but account for a large number of readmissions because of their population size.

The highest-risk groups identified in the analysis included populations associated with longer length of stay, sepsis, respiratory failure, multiple previous admissions, and ICU care. These groups represent potential populations for closer clinical attention because their observed 30-day readmission rates were substantially above the overall 12.62% benchmark.

At the same time, readmission burden was concentrated among several high-volume groups with above-average readmission rates. Secondary diagnoses accounted for 11,584 readmissions, or 81.52% of all 30-day readmissions, with a 15.97% readmission rate. Other major contributors included patients with low hemoglobin (10,756 readmissions; 15.25% readmission rate) and emergency admissions (9,859 readmissions; 15.76% readmission rate). Patients with 1–3 procedures and those treated in teaching hospitals also appeared among the leading above-average-risk contributors.

These findings indicate that the hospital should not focus exclusively on the groups with the highest readmission percentages. Risk identifies populations with greater likelihood of readmission, while burden identifies where the largest absolute number of readmissions is occurring. Both perspectives are needed to prioritize interventions.

For example, a very high-risk but relatively small diagnosis group may warrant targeted clinical attention, while a larger group with a lower-than-average readmission rate may still generate a substantial number of readmissions simply because it represents a large share of hospital discharges. This distinction is illustrated by the Risk vs. Burden analysis and is important when allocating limited intervention resources.

3.3 Deep Dive: Secondary Diagnoses

Dashboard 2: Link to Top Contributor Dashboard at Tableau Public

https://public.tableau.com/views/HospitalReadmissionConsultationProject/ClinicalDashboard?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Because secondary diagnoses represented the largest readmission burden, a focused analysis was performed to examine factors associated with readmission within this population.

The analysis identified statistically significant differences across several clinical, utilization, admission, and discharge-related factors. Important signals included ward type, admission type, primary diagnosis category, discharge type, age, length of stay, comorbidity burden, Charlson index, previous admissions, procedures, and diagnosis burden. Effect sizes were used alongside statistical significance to distinguish the magnitude of observed differences rather than treating statistical significance alone as evidence of practical importance.

These findings can help clinicians and care-management teams identify populations that may warrant closer assessment or additional transitional-care support. They should, however, be interpreted as associations rather than causal relationships. The analysis does not establish that any individual factor causes readmission.

The findings also provide operational teams with potential intervention targets. For example, elevated readmission associated with particular discharge or admission pathways may point to opportunities for improved discharge planning, follow-up, medication reconciliation, or transitional-care processes.

3.4 Predictive Modeling Adds a Patient-Level Prioritization Layer

Technical details of this predictive model: [06_03_Readmission Predictive_Modeling.ipynb](#)

Population-level analysis identifies groups that warrant attention. It does not answer the operational question:

Which individual discharged patients should receive additional attention when resources are limited?

Predictive modeling was therefore conducted using the eligible discharge population.

Model comparison and Selection

Model	Test AUC
Logistic Regression — baseline* (Selected)]	0.7545
Logistic + 34 ICD-10 indicators	0.7554
Logistic + ICD-10 categories	0.7548

Logistic + diagnosis burden	0.7544
XGBoost	0.7498
Random Forest	0.7331–0.7408

Adding 34 individual ICD-10 indicators increased AUC by only **0.0008** compared with the baseline Logistic Regression model. Given the minimal improvement and the additional complexity, the simpler baseline Logistic Regression was selected. This is an important distinction: **the selected model* was not chosen because it perfectly predicts readmission**. Its value is its combination of moderate discrimination, simplicity, and interpretability for an exploratory prioritization use case.

3.5 Risk Percentiles Can Support Resource Planning

The selected model was used to rank patients by predicted readmission risk and examine the concentration of observed readmissions across risk percentiles.

Highest-risk percentile	Patients targeted	Observed readmissions captured	Share of all test-set readmissions	Observed readmission rate
Top 5%	1,126	545	19.2%	48.4%
Top 10%	2,252	916	32.2%	40.7%
Top 15%	3,378	1,199	42.2%	35.5%
Top 20%	4,504	1,441	50.7%	32.0%
Top 25%	5,630	1,637	57.6%	29.1%

The results show that observed readmissions were concentrated among patients assigned higher predicted-risk percentiles. For example, the highest-risk 20% of patients accounted for **50.7% of observed readmissions** in the test population. This can inform planning for different intervention capacities. It does **not** establish that intervening on these patients would prevent the observed readmissions.

The model answers:

Who should we prioritize?

The future intervention program must answer:

What should we do for them, and does it work?

4. Recommendations & Intervention Planning

The findings support a **targeted transitional-care strategy** rather than a single broad intervention.

4.1 Use Risk Stratification to Support Targeting

The predictive model can be evaluated as a decision-support tool for identifying patients who may warrant additional transitional-care resources.

Potential activities include:

- enhanced discharge planning;
- medication reconciliation;
- scheduling follow-up appointments before discharge;
- pharmacist or care-manager review;
- patient education using teach-back;
- post-discharge phone or digital follow-up;
- early primary-care or outpatient follow-up; and
- coordination of transportation, medication access, or other identified barriers.

The percentile analysis can be used to examine different resource-capacity scenarios rather than establishing a single universal intervention threshold.

4.2 Address High-Burden Populations

The secondary-diagnosis population accounts for a large proportion of total readmissions and should therefore be included in future intervention planning. The hospital can examine whether targeted transitional-care workflows can be integrated into discharge pathways for these high-volume populations.

4.3 Prioritize High-Risk Clinical Populations

The analysis identified several populations with markedly elevated observed 30-day readmission rates. Patients with LOS ≥ 21 days had a 37.4% readmission rate, followed by patients with sepsis at 31.5% and respiratory failure at 30.8%. Patients with a previous admission had a 26.7% readmission rate, while those admitted to the ICU had a 25.1% rate.

These findings highlight prolonged hospitalization, severe clinical conditions, and prior utilization as important signals for transitional-care planning. Sepsis and respiratory failure are particularly notable because, despite their smaller absolute readmission counts, their readmission rates are exceptionally high. The hospital should therefore consider these populations for enhanced discharge planning and follow-up and, where resources permit, evaluate whether condition-specific transitional-care pathways would be appropriate.

4.4 Extend Monitoring Beyond the First Seven Days

Because only a small fraction of the 30-day readmissions occurred within the first 7 days, transitional-care monitoring should not be limited to the immediate post-discharge period. The hospital should consider evaluating follow-up coverage across the full 30-day period.

4.5 Measure Intervention Effectiveness Prospectively

The next stage should move from observational analysis to prospective intervention evaluation. A pilot should clearly define the target population, intervention, outcome measures, comparison approach, and evaluation period. Outcomes could include 7-day and 30-day readmission, emergency department utilization, follow-up completion, medication reconciliation completion, and patient engagement. The objective is to determine whether the intervention produces measurable improvement in patient outcomes and care processes, rather than assuming that patients identified as high risk are necessarily preventable readmissions

5. Limitations & Next Steps

Limitations

Several limitations should be considered when interpreting the findings.

Observational analysis.

The analysis identifies associations and does not establish causality.

Synthetic/project dataset.

The project is designed as a portfolio consulting case and should not be interpreted as evidence about an actual hospital population.

Potential missing information.

Insurance type contains missing values, and other variables may not capture important social, behavioral, or post-discharge factors.

Predictive performance.

An AUC of approximately 0.75 indicates useful discrimination for an exploratory model but does not imply perfect patient-level prediction.

No intervention effect estimate.

The risk model identifies patients with higher observed risk; it does not estimate which patients would benefit from intervention or how many readmissions could be prevented.

Temporal considerations.

Variables used for operational deployment would need to be verified as available at the intended prediction point and reviewed for potential leakage before implementation.

Recommended Next Steps

1. **Validate the model prospectively** using a future time period or external hospital data.
 2. **Confirm feature availability at discharge** and review all variables for leakage.
 3. **Pilot targeted transitional-care interventions** among selected risk and burden populations.
 4. **Measure intervention outcomes** against an appropriate comparison group.
 5. **Monitor model performance and equity** across relevant patient populations.
 6. **Refresh the risk model periodically** as patient populations, clinical practices, and workflows change.
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6. Technical Deliverables

The final consulting recommendations are supported by a reproducible analytical workflow documented separately in the project repository.

Conclusion

The analysis indicates that the hospital's readmission challenge is persistent and distributed across multiple patient and operational populations.

The findings also demonstrate why **readmission rate alone is insufficient for intervention planning**. Some populations have substantially elevated observed rates, while other high-volume populations account for a large share of total readmissions.

A combined approach can therefore provide greater operational value:

Population analysis identifies where the burden is concentrated.

Association analysis identifies characteristics associated with elevated observed rates.

Predictive modeling provides a mechanism for prioritizing patients when resources are constrained.

The next stage is not additional retrospective analysis alone. It is to **test targeted transitional-care interventions prospectively and measure whether they produce meaningful reductions in readmission and related outcomes**.